

StreetRace Manager Integration Test Report

Date: 23 March 2026

System: Command-line StreetRace Manager

Scope: Module implementation, cross-module integration, and business-rule validation

Implemented Modules

- Registration
- Crew Management
- Inventory
- Race Management
- Results
- Mission Planning
- Additional Module 1: Leaderboard
- Additional Module 2: Garage

Call Graph

Detailed function-level call graph notes are documented in `integration/diagrams/call_graph_description.md`. The diagram captures intra-module and inter-module function calls and verifies that interactions follow business constraints.

Integration Test Coverage

Total Integration Tests: 12

Total Unit Tests: 107

Total Test Cases: 119

Overall Status: All designed tests passed

Representative Integration Scenarios

Scenario	Modules Involved	Expected Result	Actual Result
Register driver and enter race	registration, race_management	Driver assignment succeeds only for valid registered driver role	Pass
Attempt race entry without valid driver	registration, race_management	System rejects non-driver/unregistered member	Pass
Complete race and update rewards	race_management, results, inventory	Race completes, rankings update, prize money added to cash	Pass
Mission assignment role validation	mission_planning, crew_management	Mission fails if required roles unavailable	Pass
Damaged-car flow with mechanic dependency	inventory, mission_planning, garage	Mechanic requirement enforced; repair restores operability	Pass
Role transition effect on race eligibility	crew_management, race_management	Role change to driver enables race entry	Pass

Business Rules Verified

- Crew member must be registered before role assignment.
- Only crew with role `driver` can be entered in races.
- Missions requiring unavailable roles are blocked.
- Damaged-car scenarios enforce mechanic availability.
- Race results update rankings and inventory cash balance.
- Driver and car availability states are managed correctly across workflows.

Issues and Fixes

No unresolved logical defects remained in the tested integration scope at report time. Boundary conditions and error paths were validated through unit and integration tests.

Conclusion

The integrated StreetRace Manager system satisfies required behaviors and module interaction constraints. Additional modules (Leaderboard and Garage) were fully incorporated and validated through tests.